

TECHBIG SOLUTIONS

SAP S/4HANA – ABAP Development

Complete Professional Training Syllabus

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Course Objective

A comprehensive ABAP curriculum designed for SAP S/4HANA development, covering core ABAP programming, ABAP Objects, modern Open SQL, SAP HANA development, CDS, AMDP, ABAP Development Tools (ADT), OData, SAP Fiori integration, RAP, extensibility and Clean Core principles, debugging, performance, security, testing and real-time project/support scenarios.

The syllabus intentionally combines classic ABAP skills with modern S/4HANA development. SAP's current guidance emphasizes ADT, CDS, RAP and ABAP Cloud for modern, upgrade-stable development and extensions. RAP is used for transactional Fiori applications and Web APIs, while ABAP Cloud uses released APIs and extension points.

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1. SAP S/4HANA & ABAP Platform Fundamentals

- Overview of SAP S/4HANA architecture and ABAP Platform
- ABAP application server, database and work process concepts
- Classic ABAP versus modern ABAP development
- ABAP Cloud and cloud-ready development concepts
- S/4HANA data model and simplification concepts
- Development lifecycle and transport landscape
- Developer roles, packages and software components

2. ABAP Development Tools (ADT) in Eclipse

- Installing and configuring ADT
- Connecting ADT to an SAP system
- Projects, packages and repositories
- Editors, code completion and navigation
- Creating repository objects
- Activation and syntax checks
- ABAP debugger in ADT
- ABAP source-code management and transports
- Working with released APIs and development objects

3. ABAP Programming Fundamentals

- ABAP syntax and program structure
- Data types and data objects
- Constants, literals and variables
- Structures, internal tables and work areas
- Operators and expressions
- IF, CASE, LOOP, DO and WHILE
- String processing
- Date, time and numeric calculations
- Field symbols and data references
- Inline declarations and modern syntax

4. Internal Tables & Advanced Data Handling

- Standard, sorted and hashed tables
- Table keys and secondary keys
- READ TABLE and table expressions
- LOOP AT, FILTER, VALUE and FOR expressions
- INSERT, MODIFY, DELETE and COLLECT
- Field symbols and references
- Nested internal tables

- Deep structures
- Performance considerations for large datasets

5. Modularization Techniques

- Subroutines and INCLUDE programs
- Function modules
- Function groups
- Methods and classes
- Reusable utility design
- Exception handling
- Global versus local classes
- API-oriented reusable development

6. Object-Oriented ABAP – ABAP Objects

- Classes and objects
- Public, protected and private sections
- Constructors
- Instance and static methods
- Attributes and encapsulation
- Inheritance
- Polymorphism
- Interfaces
- Abstract and final classes
- Events and event handlers
- Exception classes
- Design principles and clean OO programming

7. Modern ABAP Syntax

- Inline DATA and FIELD-SYMBOL declarations
- Constructor operators VALUE, NEW and CONV
- CORRESPONDING and REDUCE
- COND and SWITCH
- Table expressions
- String templates
- Functional expressions
- Modern LOOP and iteration expressions
- Readable and maintainable modern ABAP

8. Open SQL for SAP S/4HANA

- SELECT fundamentals
- Joins and associations concepts

- WHERE, GROUP BY, HAVING and ORDER BY
- Aggregations
- Subqueries
- Common table expressions where supported
- INSERT, UPDATE, MODIFY and DELETE
- Host variables and escaping
- Database-independent Open SQL
- SQL performance and data-volume considerations
- CDS-aware data access

9. SAP HANA & Code-to-Data Development

- SAP HANA architecture overview
- Column-store concepts
- Push-down principle
- Code-to-data versus data-to-code
- Why HANA changes ABAP development
- Efficient Open SQL
- Database access optimization
- AMDP overview
- CDS-based data modeling
- Performance analysis on HANA

10. ABAP Core Data Services (CDS)

- CDS concepts and architecture
- CDS view entities
- Data modeling with CDS
- Associations and compositions
- Annotations
- Parameters and calculated fields
- Joins and aggregations
- Authorization annotations
- Metadata extensions
- CDS projection views
- Consumption-oriented modeling
- Analytical and transactional use cases
- CDS performance and design guidelines

11. ABAP Managed Database Procedures (AMDP)

- AMDP architecture and purpose
- AMDP classes and methods
- SQLScript fundamentals

- Using AMDP with SAP HANA
- AMDP table functions
- CDS table functions overview
- Input/output parameters
- Calling AMDP from ABAP
- Performance use cases
- AMDP debugging/profiling considerations

12. Database & Application Performance

- SQL Monitor concepts
- Runtime analysis
- ABAP Trace and SAT
- ST05 SQL trace
- SQL performance patterns
- Internal table optimization
- Database access optimization
- Parallelization concepts
- Memory considerations
- Performance troubleshooting methodology

13. ABAP Dictionary / DDIC

- Domains and data elements
- Structures and table types
- Transparent tables
- Views and search helps
- Lock objects
- Database indexes
- Conversion routines
- Table maintenance concepts
- Relationships and foreign keys
- CDS versus classic DDIC modeling

14. Reports & Classical Application Development

- Executable reports
- Selection screens
- Parameters and select-options
- Variants
- ALV reports
- CL_SALV_TABLE overview
- Interactive reporting concepts
- Background execution

- Spool and job concepts
- Modern alternatives for S/4HANA applications

15. ALV & Output Development

- ALV Grid concepts
- SALV framework
- Columns, sorting and filtering
- Aggregations and totals
- Events and user commands
- Layout management
- Export and print functions
- Output design best practices

16. Function Modules, BAPIs & Business APIs

- Function modules and RFC-enabled modules
- Remote Function Calls
- BAPI concepts
- Calling standard business APIs
- Creating reusable APIs
- API error handling
- Transaction control
- Released APIs versus unreleased internal objects
- API-based S/4HANA integration

17. BDC, Migration & Legacy Techniques

- Batch Data Communication overview
- Session method
- Call Transaction
- BDC recording
- Error handling
- LSMW overview and migration context
- Legacy data migration concepts
- Modern Migration Cockpit overview
- When classic techniques should and should not be used in S/4HANA

18. Enhancements & Extensibility

- Enhancement Framework
- Customer exits
- User exits
- BAdIs
- Enhancement spots

- Implicit and explicit enhancements
- Modification-free development
- Key-user extensibility overview
- Developer extensibility
- Side-by-side extensibility
- Released extension points and APIs
- Clean Core principles

19. OData & SAP Gateway

- OData fundamentals
- REST concepts
- Gateway architecture
- OData service design
- Service implementation concepts
- Entity sets and associations
- CRUD operations
- Error handling
- Metadata and annotations
- OData V2/V4 overview
- RAP-based service exposure

20. SAP Fiori & ABAP Integration

- SAP Fiori architecture
- Fiori Launchpad concepts
- Fiori elements overview
- ABAP backend responsibilities
- CDS annotations for Fiori
- OData services
- Business catalogs and roles overview
- Value helps
- Draft concepts
- Navigation and intent-based integration

21. ABAP RESTful Application Programming Model (RAP)

- RAP architecture and business-object concept
- Managed and unmanaged implementation
- CDS root and child entities
- Behavior definitions
- Behavior implementations
- Create, update and delete operations
- Determinations

- Validations
- Actions
- Functions
- Draft handling
- Authorization control
- Feature control
- Service definition
- Service binding
- OData V4 exposure
- Fiori elements application
- Web API exposure
- RAP testing and troubleshooting

22. ABAP Cloud & Clean Core

- ABAP Cloud development model
- ABAP for Cloud Development language version
- Released SAP APIs
- Released extension points
- Developer extensibility
- Side-by-side extensions on SAP BTP
- Upgrade-stable design
- Cloud-ready application architecture
- Restrictions versus Standard ABAP
- Clean Core strategy for S/4HANA

23. Integration & Communication

- RFC and tRFC/qRFC overview
- IDoc fundamentals
- EDI concepts
- SOAP web services overview
- OData services
- REST APIs
- Event-driven integration overview
- Business events
- HTTP clients in ABAP
- JSON and XML handling
- Integration error handling

24. Forms & Output Management

- Smart Forms overview
- SAPscript overview

- Adobe Forms overview
- Form interfaces and contexts
- PDF output
- Output determination concepts
- Email and print integration
- Modern S/4HANA output considerations

25. Debugging & Troubleshooting

- ABAP debugger
- Breakpoints and watchpoints
- Conditional breakpoints
- Debugging internal tables
- Debugging classes and methods
- Update debugging
- Background-job debugging concepts
- ST22 dumps
- SM21 system log
- SLG1 application logs
- Analyzing short dumps and root causes

26. Authorization & Security for ABAP Developers

- Authorization objects
- AUTHORITY-CHECK
- Role-based access concepts
- CDS access controls (DCL)
- Data authorization in CDS
- Secure API consumption
- Input validation
- SQL injection prevention
- Sensitive-data handling
- Secure coding practices

27. Testing & Quality

- Unit testing concepts
- ABAP Unit
- Test classes and test methods
- Test doubles overview
- Integration testing
- Code Inspector
- ATC – ABAP Test Cockpit
- Static quality checks

- Performance checks
- Transport and regression testing

28. Transport & Development Lifecycle

- Workbench requests
- Customizing requests overview
- Transport organizer
- Packages and object ownership
- Release/import process
- Landscape: DEV, QAS, PRD
- Version management
- Deployment controls
- Emergency fixes
- Production support governance

29. S/4HANA Conversion & Custom Code Adaptation

- ECC to S/4HANA conversion concepts
- Custom code analysis
- Simplification items
- Deprecated objects and transactions
- CDS and released API adoption
- SQL performance remediation
- Code Inspector/ATC checks
- Data-model changes
- Custom-code remediation strategy
- Clean-core modernization

30. Real-Time ABAP Development Projects

- Project 1: Interactive S/4HANA report using modern Open SQL and ALV
- Project 2: CDS-based analytical data model
- Project 3: AMDP/HANA push-down scenario
- Project 4: OData service for a business process
- Project 5: Fiori elements application
- Project 6: RAP managed business object with CRUD
- Project 7: RAP validation, determination, action and draft
- Project 8: S/4HANA extension using released API/BAdI
- Project 9: Interface using IDoc/API/HTTP
- Project 10: Production support issue, dump analysis and performance optimization

31. Interview & Certification Preparation

- Core ABAP programming questions

- ABAP Objects and modern syntax
- Open SQL and HANA optimization
- CDS and AMDP scenarios
- RAP and ABAP Cloud questions
- Fiori/OData integration scenarios
- Enhancement and Clean Core questions
- Debugging and performance cases
- Real-time implementation scenarios
- Mock interviews and practical assessment

Recommended Training Roadmap

Phase	Coverage	Outcome
1	ABAP Foundation	Strong programming and DDIC fundamentals
2	Modern ABAP	Modern syntax, OO ABAP and Open SQL
3	HANA Development	CDS, AMDP and performance optimization
4	S/4HANA Development	OData, Fiori and extensibility
5	RAP & ABAP Cloud	Build cloud-ready transactional applications and APIs
6	Quality & Security	Testing, ATC, authorization and secure coding
7	Real-Time Projects	Build end-to-end S/4HANA applications
8	Support & Interview	Troubleshooting, performance and interview preparation

SAP Reference Alignment

SAP's current ABAP guidance emphasizes ABAP Development Tools, Core Data Services and the ABAP RESTful Application Programming Model for modern S/4HANA development. RAP supports transactional Fiori applications and Web APIs, while ABAP Cloud promotes released APIs, extension points and upgrade-stable development.

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